non-gaseous activities are contained within the plant, but small releases occur to the environment according to limits set by the regulating authorities. These releases were rather large in earlier days, but have been considerably reduced as improved procedures and equipment are coming into operation, see Table 21.10. The authorized releases in the UK in 1978 were 10^4 TBq β-emitters (excl. T) per year. The ^{137}Cs release produced ~1000 Bq/l water in the middle of the Irish Sea (~150 km SW Windscale), and could be traced as far away as in the Baltic Sea. The release into the Atlantic in 1978 was ≤10 000 TBq. Not counting releases from plants in the US, former USSR and India, it is assumed to be <1000 TBq in 1992. It should be noted that the release of radionuclides into the sea from nuclear power operations is dominated by the effluents from a few reprocessing plants. It is, however, only a small fraction of the natural radioactivity in the oceans.

21.9. TREATMENT AND DEPOSITION OF LOW AND MEDIUM LEVEL WASTES

It has become a common practice at reprocessing plants to store low level solid waste (and sometimes higher level wastes) in trenches dug from the soil. These trenches, which commonly are 5–8 m deep, are sometimes lined with concrete or simply have a gravel bottom. For this purpose dry areas are selected (deserts, when available) or isolated areas with controlled groundwater conditions with respect to water table depth, flow rate, and direction. The disposed material normally should not exceed a dose rate of 1 mGy h^{-1} at 0.3 m distance, or contain a specific activity exceeding 1 MBq kg^{-1}. However, this varies, and may be a factor of 10 higher or lower in some places. When the trench is full, it is backfilled with earth, after which the surface dose rate usually is <0.01 mGy h^{-1}. Trenches of this kind are used in the USA, the UK, France, Russia, etc., where tens of thousands of cubic meters have been disposed annually; Fig. 21.17(A).

Since some of the waste products are longlived, and the physical protection of the waste in surface trenches is poor, radioactivity may ultimately leak into the groundwater, see Ch. 22. Therefore, in many countries repositories for final deposition of solid intermediate and low level wastes have been or are being built. The Swedish repository at Forsmark for ILW and LLW from reactors, hospitals, research facilities, etc. has been in operation since 1988; Fig. 21.17(B). It is located in solid granite rock below 50 m of sea water at the Baltic Sea coast. This underground site contains large concrete silos for the ILW and storage tunnels for ILW and LLW containers. The space between the concrete walls and the rock is filled with clay (bentonite). The repository is partly operated by remote control. Fig. 21.17(C) shows Belgian plans for a mixed ILW and HLW repository located in a large clay deposit. Also in this repository, concrete and clay are the main protective barriers. Geologic barriers are discussed in §21.13.

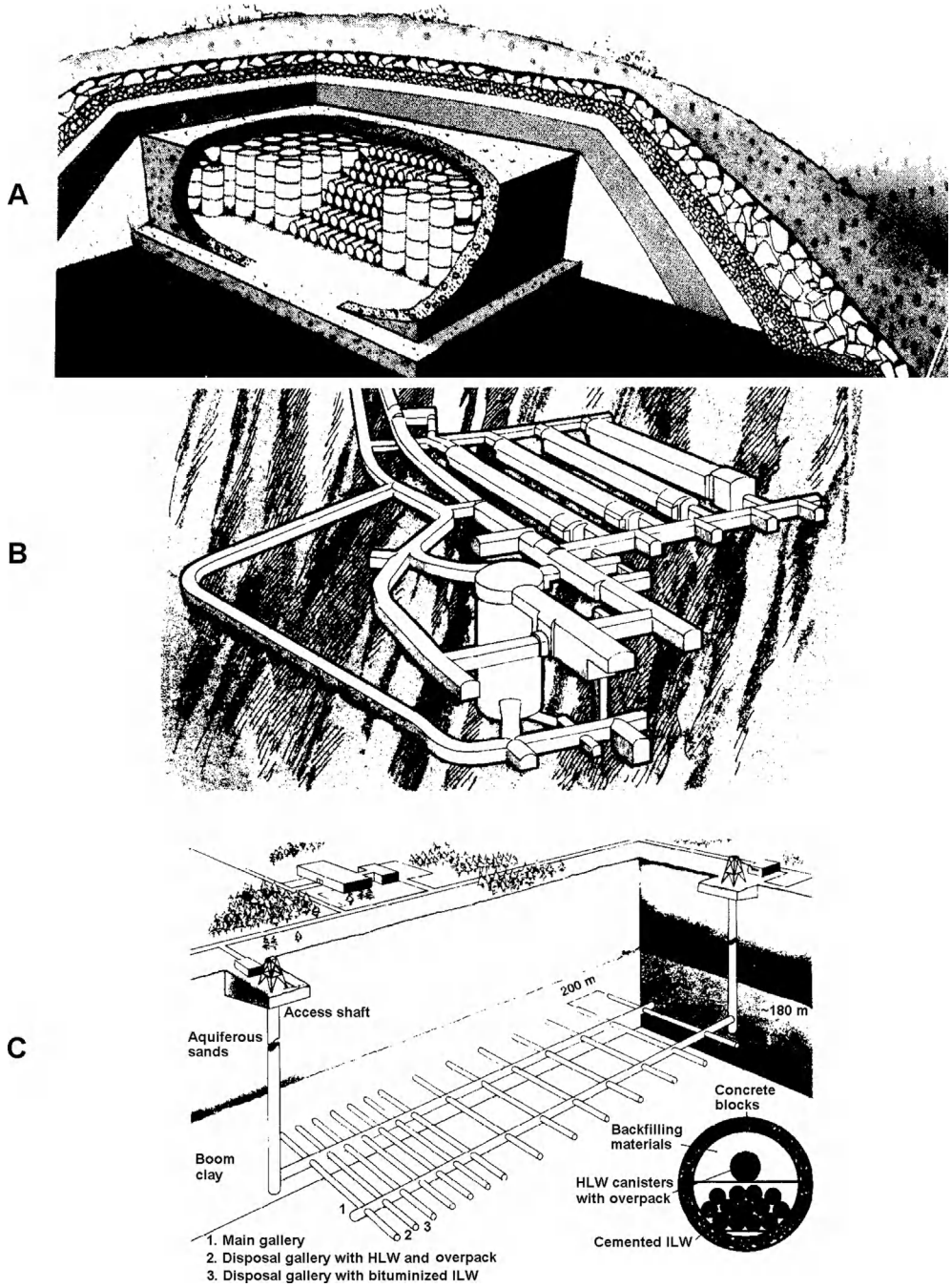


Figure 21.17 (A) Storage of solid LLW (La Hague, France); (B) Storage of LLW and ILW (Forsmark, Sweden); (C) Suggested repository for solid ILW and HLW (Belgium).

21.10. TANK STORAGE OF HIGH LEVEL LIQUID WASTES

The main part of the HLLW is aqueous raffinate from the PUREX cycle. It contains ~99.9% of the nonvolatile FPs, <0.5% of the uranium, <0.2% of the plutonium, and some corrosion products. For each ton of uranium reprocessed about 5 m^3 of HLLW is produced. This is usually concentrated to 0.5–1 m^3 for interim tank storage; specific activity is in the range 10^7 GBq m^{-3}. The amounts of various elements in the waste and their concentration in 0.5 m^3 solution is shown in Table 21.9. The HNO_3 concentration may vary within a factor of 2 depending on the concentration procedure. The metal salt concentration is ~0.5 M; it is not possible to keep the salt in solution except at high acidity. The amounts of corrosion products, phosphate, and gadolinium (or other neutron poison added) also may vary considerably. Wastes from the HTGR and FBR cycles are expected to be rather similar.

The HLLW stainless steel tanks have a volume of 50–500 m^3. They are rather elaborate (Fig. 21.18); they contain arrangements for cooling (≤65°C to reduce corrosion) and stirring, removal of radiolytic gases, and for control of liquid level, pH, and radioactivity. The tanks are usually double-walled, have heavy concrete shielding, and are often placed underground. Storage in stainless steel tanks has been used in the last 40 y without failure. The philosophy is that tank storage is only an interim procedure and

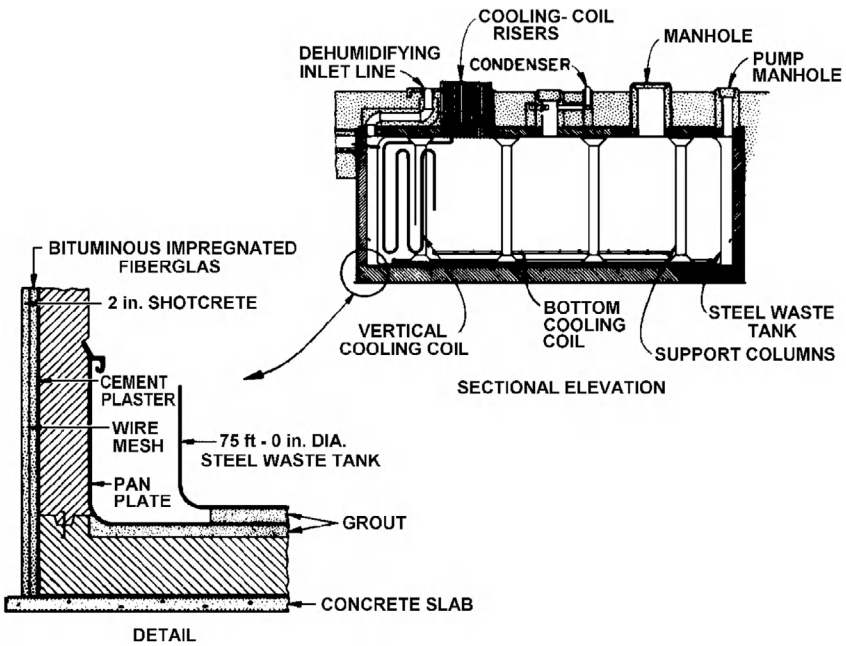


Figure 21.18 Stainless steel tank for storage of HLLW.